

Eleven models, nine corpora: the sealed suite ranks them backwards

Results

99 measurements — every model in this repository against every evaluation corpus, one fixed evaluator, one core.

Eight corpora are the sealed suite that decides gates. **The ninth**, Synthetic_PDF_Corpus_v2_1612, **is an out-of-distribution holdout**: 1,612 synthetic PDFs no training corpus resembles, held outside data/2-eval and scored only after every model was already selected.

model	labels	sealed F1	sealed P	sealed R	holdout F1	holdout P	holdout R
cascade_balanced_v4	58	0.7862	0.7188	0.8745	0.6574	0.8361	0.5416
cascade_balanced_v3	58	0.7533	0.6634	0.8856	0.6360	0.7545	0.5497
cascade_scorecard61	61	0.7299	0.6360	0.8866	0.5836	0.6606	0.5227
cascade_p90r85b1	61	0.8564	0.8721	0.8415	0.5453	0.7156	0.4405
cascade_p88r90b1	61	0.8557	0.8634	0.8483	0.5427	0.7100	0.4392
v5d_fused	61	0.7595	0.6926	0.8592	0.5296	0.6938	0.4282
cascade_p80r90	61	0.8474	0.8844	0.8141	0.5167	0.7108	0.4059
cascade_p80r70	61	0.8394	0.9165	0.7777	0.5034	0.7215	0.3866
cascade_p90r90	61	0.8464	0.9115	0.7926	0.5006	0.7156	0.3849
cascade_p88r90	61	0.8470	0.9000	0.8020	0.4992	0.7061	0.3861
v5e_fused	61	0.8373	0.9190	0.7726	0.4873	0.7308	0.3655

Sealed figures are equal-corpus over the eight. **The two 58-label rows are in italics because they are not comparable** — see below.

The sealed suite ranks the 61-label models almost exactly backwards

Across the nine models that share the 61-label catalogue:

measured on the sealed 8	Pearson r with holdout micro F1
micro recall	+0.9359
macro recall	+0.9656
micro F1	−0.6024
micro precision	−0.7793

The best sealed model (p90r85b1, F1 0.8564) is fourth on the holdout. The best sealed precision (v5e_fused, 0.9190) is **last**. The untuned baseline — worst sealed F1 of the tuned set at 0.7299 — is **best** of the 61-label models out-of-distribution.

Precision transfers; recall does not. Holdout precision spans 0.6606–0.7308 across every 61-label model, a range of 0.07. Holdout F1 spans 0.4873–0.5836 and tracks recall.

Every model loses roughly a third on unfamiliar documents

	best sealed	best holdout	drop
61-label micro F1	0.8564	0.5836	−0.27
58-label micro F1	0.7862	0.6574	−0.13

No model in this repository has been shown to work on documents unlike its training data. That is the single most important line in this report.

TL;DR

- **99 measurements**, 11 models × 9 corpora, one evaluator, recorded in 26-08-27_Experiment-Log.xlsx (99 Experiment Log rows, 5,985 per-tag rows).
- **The sealed suite is anti-correlated with generalisation.** Optimising sealed micro F1 (r = −0.60) or precision (r = −0.78) predicts *worse* out-of-distribution performance; recall predicts it almost perfectly (r = +0.97 on macro recall).
- **The untuned cascade_scorecard61 is the best 61-label model on the holdout** (0.5836), despite being the weakest tuned model on the sealed suite.
- v5e_fused **has the highest sealed precision (0.9190) and the worst holdout score (0.4873)**. The precision-first direction is the generalisation-worst direction, consistently, across all nine.
- **The 58-label models look best on the holdout and the comparison is unfair** — they are scored on a collapsed label space that merges the name subtypes, which is a materially easier task. Do not read that as evidence for the old taxonomy.
- **Everything drops ~0.27 micro F1** moving from the sealed suite to unfamiliar documents.

Date	2026-08-27
Project	projects/pii-target-8070
Run	26-08-27-1
Request	"do an eval run from all these models over all 9 eval datasets"
Scope	11 models × 9 corpora = 99 measurements
CPU budget	1 core for every published latency

Outcome	matrix recorded; no promotion — nothing has cleared a gate
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Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_scorecard61	10360_betterdataai_ner_silver_eval_10.36k				0.263		
cascade_scorecard61	10626_ai4privacy_pii_masking_eval_10.63k				0.715		
cascade_scorecard61	20000_pii_holdout_20.00k				0.739		
cascade_scorecard61	30000_pii2_eval_25.15k				0.761		
cascade_scorecard61	38937_openpii_pii_eval_38.94k				0.692		
cascade_scorecard61	4000_datax-dualjudge-evalset-1.32k						
cascade_scorecard61	5617_nemotron_eval_5.36k						
cascade_scorecard61	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_scorecard61	Synthetic_PDF_Corpus_v2_1612				0.356		
cascade_p80r70	10360_betterdataai_ner_silver_eval_10.36k				0.259		
cascade_p80r70	10626_ai4privacy_pii_masking_eval_10.63k				0.732		
cascade_p80r70	20000_pii_holdout_20.00k				0.827		
cascade_p80r70	30000_pii2_eval_25.15k				0.809		
cascade_p80r70	38937_openpii_pii_eval_38.94k				0.717		
cascade_p80r70	4000_datax-dualjudge-evalset-1.32k						
cascade_p80r70	5617_nemotron_eval_5.36k						
cascade_p80r70	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_p80r70	Synthetic_PDF_Corpus_v2_1612				0.288		
cascade_p80r90	10360_betterdataai_ner_silver_eval_10.36k				0.268		
cascade_p80r90	10626_ai4privacy_pii_masking_eval_10.63k				0.752		
cascade_p80r90	20000_pii_holdout_20.00k				0.827		
cascade_p80r90	30000_pii2_eval_25.15k				0.807		
cascade_p80r90	38937_openpii_pii_eval_38.94k				0.725		

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_p80r90	4000_datax-dualjudge-evalset-1.32k						
cascade_p80r90	5617_nemotron_eval_5.36k						
cascade_p80r90	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_p80r90	Synthetic_PDF_Corpus_v2_1612				0.295		
cascade_p88r90	10360_betterdataai_ner_silver_eval_10.36k				0.249		
cascade_p88r90	10626_ai4privacy_pii_masking_eval_10.63k				0.741		
cascade_p88r90	20000_pii_holdout_20.00k				0.828		
cascade_p88r90	30000_pii2_eval_25.15k				0.807		
cascade_p88r90	38937_openpii_pii_eval_38.94k				0.724		
cascade_p88r90	4000_datax-dualjudge-evalset-1.32k						
cascade_p88r90	5617_nemotron_eval_5.36k						
cascade_p88r90	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_p88r90	Synthetic_PDF_Corpus_v2_1612				0.286		
cascade_p90r90	10360_betterdataai_ner_silver_eval_10.36k				0.253		
cascade_p90r90	10626_ai4privacy_pii_masking_eval_10.63k				0.732		
cascade_p90r90	20000_pii_holdout_20.00k				0.825		
cascade_p90r90	30000_pii2_eval_25.15k				0.808		
cascade_p90r90	38937_openpii_pii_eval_38.94k				0.721		
cascade_p90r90	4000_datax-dualjudge-evalset-1.32k						
cascade_p90r90	5617_nemotron_eval_5.36k						
cascade_p90r90	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_p90r90	Synthetic_PDF_Corpus_v2_1612				0.282		
cascade_p90r85b1	10360_betterdataai_ner_silver_eval_10.36k				0.289		
cascade_p90r85b1	10626_ai4privacy_pii_masking_eval_10.63k				0.782		
cascade_p90r85b1	20000_pii_holdout_20.00k				0.843		
cascade_p90r85b1	30000_pii2_eval_25.15k				0.804		
cascade_p90r85b1	38937_openpii_pii_eval_38.94k				0.737		

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_p90r85b1	4000_datax-dualjudge-evalset-1.32k						
cascade_p90r85b1	5617_nemotron_eval_5.36k						
cascade_p90r85b1	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_p90r85b1	Synthetic_PDF_Corpus_v2_1612				0.320		
cascade_p88r90b1	10360_betterdataai_ner_silver_eval_10.36k				0.287		
cascade_p88r90b1	10626_ai4privacy_pii_masking_eval_10.63k				0.785		
cascade_p88r90b1	20000_pii_holdout_20.00k				0.841		
cascade_p88r90b1	30000_pii2_eval_25.15k				0.804		
cascade_p88r90b1	38937_openpii_pii_eval_38.94k				0.737		
cascade_p88r90b1	4000_datax-dualjudge-evalset-1.32k						
cascade_p88r90b1	5617_nemotron_eval_5.36k						
cascade_p88r90b1	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_p88r90b1	Synthetic_PDF_Corpus_v2_1612				0.321		
v5d_fused	10360_betterdataai_ner_silver_eval_10.36k				0.245		
v5d_fused	10626_ai4privacy_pii_masking_eval_10.63k				0.711		
v5d_fused	20000_pii_holdout_20.00k				0.727		
v5d_fused	30000_pii2_eval_25.15k				0.776		
v5d_fused	38937_openpii_pii_eval_38.94k				0.694		
v5d_fused	4000_datax-dualjudge-evalset-1.32k						
v5d_fused	5617_nemotron_eval_5.36k						
v5d_fused	6589_govdocs2-dualjudge-eval20-3.53k						
v5d_fused	Synthetic_PDF_Corpus_v2_1612				0.262		
v5e_fused	10360_betterdataai_ner_silver_eval_10.36k				0.234		
v5e_fused	10626_ai4privacy_pii_masking_eval_10.63k				0.716		
v5e_fused	20000_pii_holdout_20.00k				0.790		
v5e_fused	30000_pii2_eval_25.15k				0.802		
v5e_fused	38937_openpii_pii_eval_38.94k				0.717		

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
v5e_fused	4000_datax-dualjudge-evalset-1.32k						
v5e_fused	5617_nemotron_eval_5.36k						
v5e_fused	6589_govdocs2-dualjudge-eval20-3.53k						
v5e_fused	Synthetic_PDF_Corpus_v2_1612				0.237		
cascade_balanced_v4	10360_betterdataai_ner_silver_eval_10.36k				0.290		
cascade_balanced_v4	10626_ai4privacy_pii_masking_eval_10.63k				0.764		
cascade_balanced_v4	20000_pii_holdout_20.00k				0.727		
cascade_balanced_v4	30000_pii2_eval_25.15k				0.764		
cascade_balanced_v4	38937_openpii_pii_eval_38.94k				0.674		
cascade_balanced_v4	4000_datax-dualjudge-evalset-1.32k						
cascade_balanced_v4	5617_nemotron_eval_5.36k						
cascade_balanced_v4	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_balanced_v4	Synthetic_PDF_Corpus_v2_1612				0.367		
cascade_balanced_v3	10360_betterdataai_ner_silver_eval_10.36k				0.289		
cascade_balanced_v3	10626_ai4privacy_pii_masking_eval_10.63k				0.742		
cascade_balanced_v3	20000_pii_holdout_20.00k				0.720		
cascade_balanced_v3	30000_pii2_eval_25.15k				0.760		
cascade_balanced_v3	38937_openpii_pii_eval_38.94k				0.669		
cascade_balanced_v3	4000_datax-dualjudge-evalset-1.32k						
cascade_balanced_v3	5617_nemotron_eval_5.36k						
cascade_balanced_v3	6589_govdocs2-dualjudge-eval20-3.53k						
cascade_balanced_v3	Synthetic_PDF_Corpus_v2_1612				0.378		

Measured 2026-08-27 · n=1,612, n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.